The state of Qi, fourth century BC



淳于髡一日而見七人於宣王。

In one day, Chunyu Kun presented seven men to King Xuan.

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



Strategies of the Warring States

Zhanguo ce 戰國策, edited by Liu Xiang in the 1st century BC

Stories of the politics and wars of the Chinese states, 5th to 3rd century BC.

King Xuan of Qi gathered hundreds of scholars in his capital. One of them, Chunyu Kun, known for his wit, presented new scholars to the king.

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain

King Xuan of Qi

reigned 319 to 301 BC

"Come. I have heard that one scholar in a thousand li is as if they stood shoulder to shoulder, and one sage in a hundred generations as if they came heel after heel. Now you present seven scholars in one morning: are scholars not plentiful, then?"

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer

Chunyu Kun

c. 386 to 310 BC

"Not so. Birds with the same wings live together; beasts with the same feet walk together. Look for chaihu and jiegeng in the marshes, and in generations you will not find one; go to the north slopes of the mountains 罩黍 and 梁父, and you load them by the cartful."

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer



Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain

Weekend 2

Representation learning

We organize data so that similar data points cluster together.

For each data point, we compute a representation, its embedding: a list of numbers that captures its most characteristic features.

Data points that belong together get similar embeddings.

Embeddings are one of the main drivers of AI today.

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain